HW # 5 Solutions

P.1 (Simon 5.2)

(a) for Hydrogen atom we have $E_n = -\frac{m_e e^4}{2(4\pi\varepsilon_0)^2\hbar^2}\frac{1}{n^2}$

$$\simeq -13.6\ \frac{1}{n^2}\ eV$$

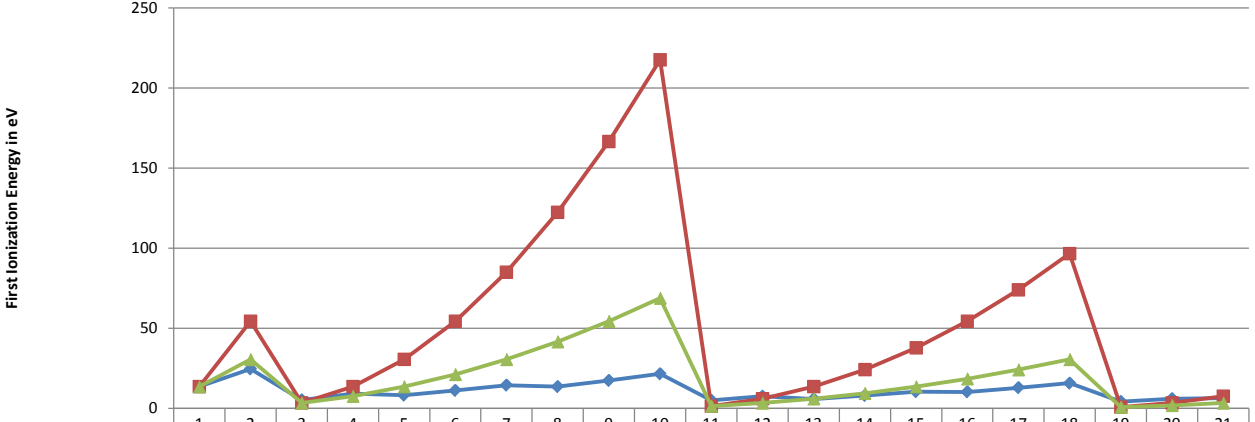
Since we have e^4 in E_n expression for atomic number Z the expression would be

$$E_n \simeq -13.6\ \frac{Z^2}{n^2}\ eV$$

(b) Approx a: here we think about the total charge in the inner shell which is of course $e(Z - N_{inside})$

Approx b: consider also screening due to the electrons in the same shell as well. Think of them as half inside and half outside, so the effect would be $\left(Z - N_{inside} - \frac{N_{same} - 1}{2}\right)$

* for higher atomic numbers outside shells are really mixed up



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Actuall Value	13.59	24.58	5.391	9.322	8.298	11.26	14.53	13.61	17.42	21.56	5.139	7.646	5.985	8.151	10.48	10.36	12.96	15.76	4.340	6.113	6.561
Approximation a	13.59	54.39	3.399	13.59	30.59	54.39	84.99	122.3	166.5	217.5	1.510	6.043	13.59	24.17	37.77	54.39	74.03	96.7	0.849	3.399	7.649
Approximation b	13.59	30.59	3.399	7.649	13.59	21.24	30.59	41.64	54.39	68.84	1.510	3.399	6.043	9.443	13.59	18.50	24.17	30.59	0.849	1.912	3.399

P.2 (Simon 6.2)

[a] $$E = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\sum_{mn} \varphi_m^* H_{mn} \varphi_n}{\sum_m |\varphi_m|^2}$$

$$\varphi_n = \varphi_n' + i\varphi_n'' \quad \& \quad \varphi_n^* = \varphi_n' - i\varphi_n''$$

$$\frac{\partial}{\partial \varphi_n'} \sum_{mn} \varphi_m^* H_{mn} \varphi_n = 2\,\mathrm{Re} \sum_m H_{nm} \varphi_m$$

$$\frac{\partial}{\partial \varphi_n''} \sum_{mn} \varphi_m^* H_{mn} \varphi_n = 2\,\mathrm{Im} \sum_m H_{nm} \varphi_m$$

$$\frac{\partial}{\partial \varphi_n'} \sum_m |\varphi_m|^2 = 2\varphi_n' \; ; \; \frac{\partial}{\partial \varphi_n''} \sum_m |\varphi_m|^2 = 2\varphi_n''$$

$$\frac{\partial E}{\partial \varphi_n'} = 0 \rightarrow 2\,\mathrm{Re} \sum_m H_{nm} \varphi_m \times \sum_m |\varphi_m|^2 = 2\varphi_n' \sum_{m\ell} \varphi_m^* H_{m\ell} \varphi_\ell$$

$$\frac{\partial E}{\partial \varphi_n''} = 0 \rightarrow 2\,\mathrm{Im} \sum_m H_{nm} \varphi_m \times \sum_m |\varphi_m|^2 = 2\varphi_n'' \sum_{m\ell} \varphi_m^* H_{m\ell} \varphi_\ell$$

but $\dfrac{\sum_{m\ell} \varphi_m^* H_{m\ell} \varphi_\ell}{\sum_m |\varphi_m|^2} = E$ Combining both eqs we get

$$\sum_m H_{nm} \varphi_m = E \varphi_n \quad \text{OR in Matrix form} \quad H\vec{\varphi} = E\vec{\varphi}$$

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[b] Since both atoms are the same and everything is symmetric upon exchanging 1 & 2 $\langle 1 | V_2 | 1 \rangle = \langle 2 | V_1 | 2 \rangle$

and $\langle 1 | V_2 | 2 \rangle = \langle 1 | V_1 | 2 \rangle$

$$H_{11} = \langle 1 | K + V_1 + V_2 | 1 \rangle = \epsilon + \langle 1 | V_2 | 1 \rangle = \epsilon + V_{cross}$$

$$H_{22} = \langle 2 | K + V_1 + V_2 | 2 \rangle = \epsilon + \langle 2 | V_1 | 2 \rangle = \epsilon + V_{cross}$$

$$H_{12} = \langle 1 | K + V_1 + V_2 | 2 \rangle = \langle 1 | V_1 | 2 \rangle = -t$$

$$H_{21} = \langle 2 | K + V_1 + V_2 | 1 \rangle = \langle 2 | V_2 | 1 \rangle = -t^*$$

$$\rightarrow H = \begin{pmatrix} \epsilon + V_{cross} & -t \\ -t^* & \epsilon + V_{cross} \end{pmatrix}$$

$$\det(H - EI) = 0 \quad \rightarrow (\epsilon + V_{cross} - E)^2 = |t|^2$$

$$\rightarrow E = \epsilon + V_{cross} \pm |t|$$

* $V_{cross} = \langle 1 | V_2 | 1 \rangle$ is significant when $\underline{\underline{r}}$ is close to R_1 and in that region $V_2 \sim V(R_1 - R_2)$ which is just about the Coulomb repulsion of two nuclei. of Course when two nuclei are too close one can't think about almost intact orbitals around each.

P.3 (Simon 6.6)

$$H_1 = \frac{e^2}{4\pi\varepsilon_0}\left[\frac{1}{|R|} + \frac{1}{|R - r_1 + r_2|} - \frac{1}{|R + r_2|} - \frac{1}{|R - r_1|}\right]$$

$$|R + r_2|^{-1} = [R^2 + r_2^2 + 2Rx_2]^{-1/2} = \frac{1}{R}\left[1 + \left(\frac{r_2}{R}\right)^2 + \frac{x_2}{R}\right]^{-1/2}$$

using $(1+\varepsilon)^{-1/2} \sim 1 - \frac{\varepsilon}{2} + \frac{3}{8}\varepsilon^2 + O(\varepsilon^3)$ up to $O(1/R^4)$

$$|R + r_2|^{-1} \sim \frac{1}{R}\left[1 - \frac{x_2}{R} + \frac{1}{2R^2}(3x_2^2 - r_2^2)\right]$$

As one can expect $\frac{1}{R}$ & $\frac{1}{R^2}$ terms vanish and we get

$$\frac{1}{|R|} + \frac{1}{|R - r_1 + r_2|} - \frac{1}{|R + r_2|} - \frac{1}{|R - r_1|} \sim \frac{1}{2R^3}\left[r_1^2 + r_2^2 - 3(x_1^2 + x_2^2) + 3(x_2 - x_1)^2 - |r_2 - r_1|^2\right]$$

$$= \frac{1}{R^3}\left[r_1 \cdot r_2 - 3x_1x_2\right]$$

$$\Rightarrow H_1 \sim \frac{e^2}{4\pi\varepsilon_0 R^3}\left[r_1 \cdot r_2 - 3x_1x_2\right] = \boxed{\frac{e^2}{4\pi\varepsilon_0 R^3}\left[z_1z_2 + y_1y_2 - 2x_1x_2\right]}$$

$$* \; E_{dipole} = \frac{1}{4\pi\varepsilon_0 R^3}\left(3(\vec{P}\cdot\hat{R})\hat{R} - \vec{P}\right)$$

$$\mathcal{E}_{d\text{-}d} = -P \cdot E = \frac{1}{4\pi\varepsilon_0 R^3}\left(\vec{P}_1 - 3(\vec{P}_1 \cdot \hat{R})\hat{R}\right) \cdot \vec{P}_2$$ then with $\vec{p} = e\vec{r}$

$$= \boxed{\frac{e^2}{4\pi\varepsilon_0 R^3}\left[\vec{r}_1 \cdot \vec{r}_2 - 3x_1x_2\right]}$$

(b) Perturbation Theory

* first order: $\delta E^{(1)} \propto \langle 1,0,0;1,0,0 | z_1 z_2 + y_1 y_2 - 2x_1 x_2 | 1,00;100 \rangle$

but $|1,0,0\rangle$ is even w.r.t all axes (x,y,z) therefore average over any of $x_1, x_2, y_1, y_2, z_1, z_2$ would be zero

So we have to go to the second order

* Second order: $\delta E^{(2)} = \sum_{\substack{n_1 n_2 \\ \ell_1 \ell_2 \\ m_1 m_2}} \frac{|\langle 100,100 | H_1 | n_1 \ell_1 m_1 ; n_2 \ell_2 m_2 \rangle|^2}{E_{0,0} - E_{n_1,n_2}}$ (I)

Since $E_{n_1,n_2} > E_0$ and we square in numerator $\delta E^{(2)} < 0$

Which means the effective force must be attrative

(c) again in (I) if any of n_1 or n_2 is one then that wave function is even and gives us zero from averaging any of x_i, y_i or z_i, therefore the first non-zero term comes from $E_{2,2}$.

(II) $$\frac{\Sigma}{|E_{0,0}|} < |\delta E| < \frac{\Sigma}{|E_{0,0} - E_{2,2}|} \quad \text{with } \Sigma = \sum_{\substack{n_1 n_2 \\ \ell_1 \ell_2 \\ m_1 m_2}} |\langle 100;100 | H_1 | n_1 \ell_1 m_1 ; n_2 \ell_2 m_2 \rangle|^2$$

$$\Sigma = \sum \langle 100,100 | H_1 | \overbrace{n_1 \ell_1 m_1, n_2 \ell_2 m_2 \rangle \langle n_1 \ell_1 m_1, n_2 \ell_2 m_2}^{\mathbb{1}} | H_1 | 100, 100 \rangle$$

$$= \sum \langle 100,100 | H_1^2 | 100,100 \rangle$$

$H_1^2 = \left(\frac{e^2}{4\pi\varepsilon_0 R^3}\right)^2 (z_1 z_2 + y_1 y_2 - 2x_1 x_2)^2$ and in the average of H_1^2

over the ground state only even terms survive, therefore

(III) $$\Sigma = \left(\frac{e^2}{4\pi\varepsilon_0 R^3}\right)^2 \sum \langle 100,100 | z_1^2 z_2^2 + y_1^2 y_2^2 + 4x_1^2 x_2^2 | 100,100 \rangle = \left(\frac{e^2}{4\pi\varepsilon_0 R^3}\right)^2 6a_0^4$$

Where we used $\langle 100 | x^2 | 100 \rangle = a_0^2$

Also noting $|E_{00}| = \dfrac{e^2}{4\pi\varepsilon_0 a_0}$ & $|E_{22} - E_{00}| = \frac{3}{4} |E_{00}|$

Using the result in III we get the result from II:

$$\boxed{\frac{6 e^2 a_0^5}{4\pi\varepsilon_0 R^6} < |\delta E| < \frac{8 e^2 a_0^5}{4\pi\varepsilon_0 R^6}}$$